Northwind Renewal Project

Selena Nguyen · Data Analyst Portfolio Project · 2026

BigQuery · Looker Studio · SQL · Data Analysis



Project Overview

Northwind Science

A company that sells science labs to school districts





Renewal Rates

Fictional Company · Synthetic Data

Project Overview

1. Are renewals at risk when account owners change?
2. Does low product utilization impact renewals?
3. Which accounts are at risk of not renewing?

Data Sources & ETL Pipeline

Medallion Architecture

Source Systems · Raw

- **Salesforce**
- **NetSuite**
- **Qualtrics**

BigQuery · Data Warehouse

BRONZE
Raw, as-landed from each source

SILVER
Cleaned, conformed & deduplicated

GOLD
Business-ready, analysis-ready tables

Tables: current_accounts · current_opportunities · dim_accounts
dim_dates · dim_users · fact_product_usage

Looker Studio

Reporting Layer

Reporting & Visualization

Dashboards for
CS & Sales

Skills: Presentation & Reporting

Q1: Are Renewals at Risk When Account Owners Change?

Q1

BigQuery SQL

```sql
with owner_history as (
  select account_id
    , date(record_valid_from) as owner_change_date
    , account_owner_id as current_owner_id
    , lag(account_owner_id) over (partition by account_id order by record_valid_from) as prior_owner_id
  from `project-d4d9e3db-6339-426d-80c.sales_mart.dim_accounts`
  where is_deleted = false
)

, owner_changes as (
  select account_id
    , owner_change_date
    , current_owner_id
    , prior_owner_id
  from owner_history
  where prior_owner_id is not null
    and prior_owner_id != current_owner_id
)

, closed_renewals as (
  select *
  from `project-d4d9e3db-6339-426d-80c.sales_mart.current_opportunities`
  where revenue_type in ('Renewal', 'Expansion Renewal')
    and stage_name in ('Closed Won', 'Closed Lost')
)

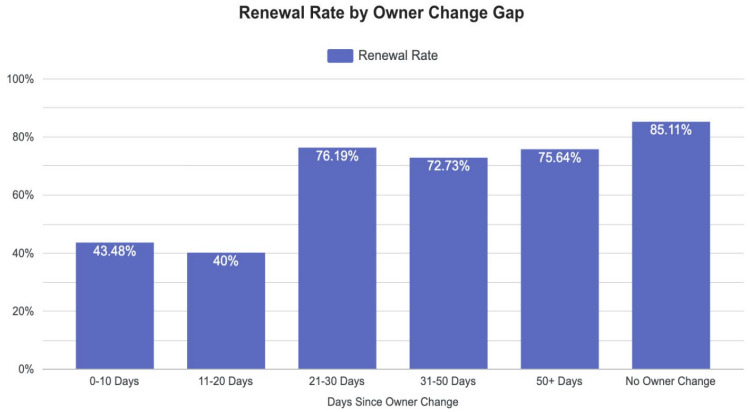
, last_change_before_close as (
  select r.opportunity_id
    , r.stage_name
    , r.available_to_renew
    , r.close_date
    , max(c.owner_change_date) as last_owner_change_date
    , date_diff(r.close_date, max(c.owner_change_date), day) as days_since_owner_change
  from closed_renewals as r
  left join owner_changes as c
    on c.account_id = r.account_id
    and c.owner_change_date <= r.close_date
  group by r.opportunity_id
  , r.stage_name
  , r.available_to_renew
  , r.close_date
)

select
  case when days_since_owner_change <= 10 then '0-10 Days'
       when days_since_owner_change <= 20 then '11-20 Days'
       when days_since_owner_change <= 30 then '21-30 Days'
       when days_since_owner_change <= 50 then '31-50 Days'
       when days_since_owner_change is null then 'No Owner Change'
       else '50+ Days'
       end as gap_bucket
  , count(*) as renewals
  , countif(stage_name = 'Closed Won') as won_renewals
  , safe_divide(countif(stage_name = 'Closed Won'), count(*)) as win_rate
  , sum(if(stage_name = 'Closed Lost', available_to_renew, 0)) as lost_renewal_value
from last_change_before_close
group by gap_bucket
order by gap_bucket
```

Q1: Findings

Q1

Q1: Are Renewals at Risk When Account Owners Change?



of Renewals by Owner Change Gap

	Days Since Owner Change ▲	Won Renewals	Total Renewals	Renewal Rate
1.	0-10 Days	10	23	43.48%
2.	11-20 Days	6	15	40%
3.	21-30 Days	16	21	76.19%
4.	31-50 Days	24	33	72.73%
5.	50+ Days	118	156	75.64%
6.	No Owner Change	40	47	85.11%

1 - 6 / 6

Key Insights

- Renewal rates are most impacted when an account owner changes **within 20 days**
- Renewal rates **drop about 35%** if an account owner changes within 11-20 days compared to 21-30 days beforehand
- **No owner changes** show the highest renewal rate

Q2: Does Low Product Utilization Impact Renewals?

Q2

BigQuery SQL

```sql
with closed_renewals as (
  select *
  from `project-d4d9e3db-6339-426d-80c.sales_mart.current_opportunities`
  where revenue_type in ('Renewal', 'Expansion Renewal')
    and stage_name in ('Closed Won', 'Closed Lost')
)

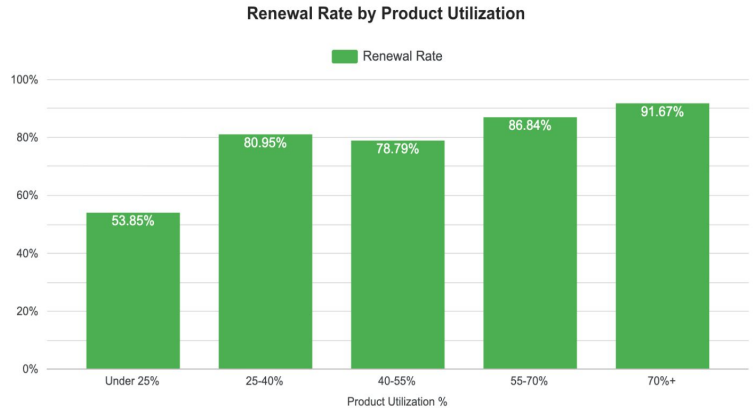
, usage as (
  select r.opportunity_id
  , safe_divide(sum(u.active_users), sum(u.licensed_seats)) as utilization_90days
  from closed_renewals as r
  inner join `project-d4d9e3db-6339-426d-80c.sales_mart.fact_product_usage` as u
    on u.account_id = r.account_id
    and u.usage_month between date_sub(date_trunc(r.close_date, month), interval 3 month)
                          and date_sub(date_trunc(r.close_date,month), interval 1 month)
  group by r.opportunity_id
)

select
  case when u.utilization_90days < 0.25 then  'Under 25%'
       when u.utilization_90days < 0.40 then '25-40%'
       when u.utilization_90days < 0.55 then '40-55%'
       when u.utilization_90days < 0.70 then '55-70%'
       else '70%+'
       end as utilization_bucket
  , count(*) as renewals
  ,countif(r.stage_name = 'Closed Won') as won_renewals
  ,safe_divide(countif(r.stage_name = 'Closed Won'), count(*)) as win_rate
from closed_renewals as r
inner join usage as u
  on u.opportunity_id = r.opportunity_id
group by utilization_bucket
order by utilization_bucket
```

Q2: Findings

Q2

Q2: Does Low Product Utilization Impact Renewals?



of Renewals by Product Utilization

	Product Utilization %	Won Renewals	Total Renewals	Renewal Rate
1.	Under 25%	56	104	53.85%
2.	25-40%	51	63	80.95%
3.	40-55%	52	66	78.79%
4.	55-70%	33	38	86.84%
5.	70%+	22	24	91.67%

1 - 5 / 5

Key Insights

- Customers who have a product utilization score of **under 25%** have the lowest renewal rate
- Renewal rates remain relatively consistent across those who have a product utilization score between 25-70%
- Customers who have a product utilization score of **70% or higher** have the highest renewal rate

Q3: Which Accounts Are At Risk of Not Renewing?

Q3

BigQuery SQL

```sql
-- Since I'm working with demo data, this is the equivalent of "current_date" (today's date)
DECLARE as_of DATE DEFAULT DATE '2026-10-20';

with open_renewals as (
  select *
  from `project-d4d9e3db-6339-426d-80c.sales_mart.current_opportunities`
  where revenue_type in ('Renewal', 'Expansion Renewal')
    and stage_name not in ('Closed Won', 'Closed Lost')
)

, recent_usage as (
  select account_id
    , safe_divide(sum(active_users),sum(licensed_seats)) as utilization_90days
  from `project-d4d9e3db-6339-426d-80c.sales_mart.fact_product_usage`
  where usage_month between date_sub(date_trunc(as_of, month), interval 3 month)
                        and date_sub(date_trunc(as_of, month), interval 1 month)
  group by account_id
)

, owner_history as (
  select account_id
    , date(record_valid_from) as owner_change_date
    , account_owner_id as current_owner_id
    , lag(account_owner_id) over (partition by account_id order by record_valid_from) as prior_owner_id
  from `project-d4d9e3db-6339-426d-80c.sales_mart.dim_accounts`
  where is_deleted = false
)

, owner_changes as (
  select account_id
    , max(owner_change_date) as last_owner_change_date
  from owner_history
  where prior_owner_id is not null
    and prior_owner_id != current_owner_id
  group by account_id
)

select a.account_name
, u.user_full_name as owner
, o.close_date as expected_close_date
, o.available_to_renew
, round(r.utilization_90days, 2) as utilization_90days
, date_diff(as_of, c.last_owner_change_date, day) as days_since_owner_change
, ifnull(r.utilization_90days < 0.25, false) as low_usage
, ifnull(date_diff(as_of, c.last_owner_change_date, day) <= 30, false) as recent_owner_change
from open_renewals as o
inner join `project-d4d9e3db-6339-426d-80c.sales_mart.current_accounts` as a
  on a.account_id = o.account_id
left join `project-d4d9e3db-6339-426d-80c.sales_mart.dim_users` as u
  on u.user_id = o.owner_id
left join recent_usage as r
  on r.account_id = o.account_id
left join owner_changes as c
  on c.account_id = o.account_id
where ifnull(r.utilization_90days < 0.25, false) = true
  or ifnull(date_diff(as_of, c.last_owner_change_date, day) <= 30, false) = true
order by low_usage desc -- outputs true
, recent_owner_change desc -- flagging accounts at risk of low usage and ownership changes recently

-- o.available_to_renew desc -- highest amount of revenue
-- o.expected_close_date desc -- prioritizing upcoming renewals
```

Q3: Findings

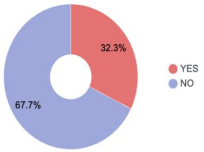
Q3

Q3: Which Accounts are at Risk of Not Renewing?

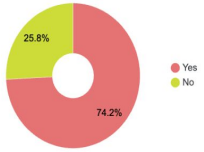
Total Renewals
31

Total Amount
$8,250,700

Renewals with Owner Change



Renewals with Low Usage



At-Risk Renewals

	Account	Owner	Expected Close Date	Risk Flag	Amount	Days Since Owner Change	Product Utilization % Last 90 Days
1.	Cedar Grove Regional ESA	Yuki Tanaka	Feb 12, 2027	Both	$552,300	4	16%
2.	Cedar Grove Regional ESA	Yuki Tanaka	Dec 24, 2026	Both	$360,700	4	16%
3.	Willowbrook County Schools	Aisha Mohammed	Nov 14, 2026	Low Usage	$321,600	51	1%
4.	Summit Ridge Education Service District	Wei Zhang	Jan 4, 2027	Low Usage	$101,200	96	2%
5.	Cedar Grove County Schools	Tyler Brooks	Nov 9, 2026	Low Usage	$124,700	226	2%
6.	Summit Ridge Education Service District	Priya Raman	Oct 26, 2026	Low Usage	$149,500	96	2%
7.	Copperfield Community Schools	Fatima Haddad	Oct 25, 2026	Low Usage	$348,500	64	5%
8.	Copperfield Community Schools	Wei Zhang	Jul 16, 2026	Low Usage	$453,700	64	5%
9.	Copperfield Community Schools	Fatima Haddad	Nov 26, 2026	Low Usage	$128,300	64	5%
10.	Brookstone Community Schools	Tyler Brooks	Nov 5, 2026	Low Usage	$204,000	137	6%
11.	Blue Ridge Education Co-op	Noah Patel	Dec 1, 2026	Low Usage	$101,600	196	7%
12.	Blue Ridge Education Co-op	Noah Patel	Dec 26, 2026	Low Usage	$350,900	196	7%
13.	Blue Ridge Education Co-op	Noah Patel	Dec 2, 2026	Low Usage	$1,013,100	196	7%
14.	Blue Ridge Education Co-op	Noah Patel	Nov 19, 2026	Low Usage	$1,079,200	196	7%
15.	Brookstone Consolidated Schools	Greg Kowalski	Oct 26, 2026	Low Usage	$208,400	73	8%

1 - 31 / 31

Key Insights

- There are **31 accounts worth $8.2M at risk** of not renewing
- **2 accounts have both** low usage and an account owner change, putting them at the highest risk of churning
- Among the 31 accounts, **32.3%** have had an owner change and **74.2%** have low product usage

Key Takeaways

1. Renewal rates are most impacted when an account owner changes **within 20 days**

2. Customers who have a product utilization score of **under 25%** have the lowest renewal rate

3. There are **31 accounts worth $8.2M at risk** of not renewing

Recommendations

1. Minimize account owner changes within **20 days** of a renewal close date

2. Flag accounts that have a product utilization score of **under 25%** and offer additional support (check-ins, training, review plan subscription and features)

3. Have the CS team focus their priority on accounts that have upcoming renewals that are at **high-risk of churning**

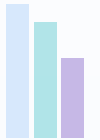
Reflection

1. Rebuild this as a reusable dbt model to keep the logic version-controlled, reusable, and accessible to the rest of the team

2. Pulling a list of specific accounts that have had an account owner change rather than only viewing the buckets

3. 3 month check-in with the team to see if renewal rates and revenue have increased after implementing business recommendations

Thank You!



Questions or feedback? Let's connect!

- **GitHub Repository**

  https://github.com/selenanguyen980/northwind_renewal_project